

Let

$$A = \begin{bmatrix} 0 & 2 & 3 & -4 & 1 \\ 0 & 0 & 2 & 3 & 4 \\ \textcircled{2} & 2 & -5 & 2 & 4 \\ 2 & 0 & -6 & 9 & 7 \end{bmatrix}.$$

Pivot column $\longrightarrow$
Pivot $\nwarrow$

We interchange the first and third rows of A , obtaining

$$B = A_{r_1 \leftrightarrow r_3} = \begin{bmatrix} \textcircled{2} & 2 & -5 & 2 & 4 \\ 0 & 0 & 2 & 3 & 4 \\ 0 & 2 & 3 & -4 & 1 \\ 2 & 0 & -6 & 9 & 7 \end{bmatrix}.$$

Multiply the first row of B by the reciprocal of the pivot, that is, by $\frac{1}{b_{11}} = \frac{1}{2}$, to obtain

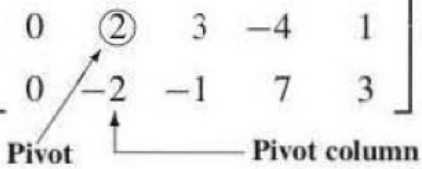
$$C = B_{\frac{1}{2}r_1 \rightarrow r_1} = \begin{bmatrix} 1 & 1 & -\frac{5}{2} & 1 & 2 \\ 0 & 0 & 2 & 3 & 4 \\ 0 & 2 & 3 & -4 & 1 \\ 2 & 0 & -6 & 9 & 7 \end{bmatrix}.$$

Add (-2) times the first row of C to the fourth row of C to produce a matrix D in which the only nonzero entry in the pivot column is $d_{11} = 1$:

$$D = C_{-2r_1 + r_4 \rightarrow r_4} = \begin{bmatrix} 1 & 1 & -\frac{5}{2} & 1 & 2 \\ 0 & 0 & 2 & 3 & 4 \\ 0 & 2 & 3 & -4 & 1 \\ 0 & -2 & -1 & 7 & 3 \end{bmatrix}.$$

Identify A_1 as the submatrix of D obtained by deleting the first row of D : Do not erase the first row of D . Repeat the preceding steps with A_1 instead of A .

$$A_1 = \begin{bmatrix} 1 & 1 & -\frac{5}{2} & 1 & 2 \\ 0 & 0 & 2 & 3 & 4 \\ 0 & \textcircled{2} & 3 & -4 & 1 \\ 0 & -2 & -1 & 7 & 3 \end{bmatrix}.$$



Do $(A_1)_{r_1 \leftrightarrow r_3}$ to obtain B_1 .

$$B_1 = \begin{bmatrix} 1 & 1 & -\frac{5}{2} & 1 & 2 \\ 0 & 2 & 3 & -4 & 1 \\ 0 & 0 & 2 & 3 & 4 \\ 0 & -2 & -1 & 7 & 3 \end{bmatrix}.$$

Do $(B_1)_{\frac{1}{2}r_2 \rightarrow r_2}$ to obtain C_1 .

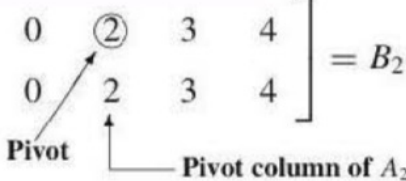
$$C_1 = \begin{bmatrix} 1 & 1 & -\frac{5}{2} & 1 & 2 \\ 0 & 1 & \frac{3}{2} & -2 & \frac{1}{2} \\ 0 & 0 & 2 & 3 & 4 \\ 0 & -2 & -1 & 7 & 3 \end{bmatrix}.$$

Do $(C_1)_{2r_1 + r_3 \rightarrow r_3}$ to obtain D_1 .

$$D_1 = \begin{bmatrix} 1 & 1 & -\frac{5}{2} & 1 & 2 \\ 0 & 1 & \frac{3}{2} & -2 & \frac{1}{2} \\ 0 & 0 & 2 & 3 & 4 \\ 0 & 0 & 2 & 3 & 4 \end{bmatrix}.$$

Deleting the first row of D_1 yields the matrix A_2 . We repeat the procedure with A_2 instead of A . No rows of A_2 have to be interchanged.

$$A_2 = \begin{bmatrix} 1 & 1 & -\frac{5}{2} & 1 & 2 \\ 0 & 1 & \frac{3}{2} & -2 & \frac{1}{2} \\ 0 & 0 & 2 & 3 & 4 \\ 0 & 0 & 2 & 3 & 4 \end{bmatrix} = B_2. \quad \text{Do } (B_2)_{\frac{1}{2}r_1 \rightarrow r_1} \text{ to obtain } C_2.$$



$$C_2 = \begin{bmatrix} 1 & 1 & -\frac{5}{2} & 1 & 2 \\ 0 & 1 & \frac{3}{2} & -2 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{3}{2} & 2 \\ 0 & 0 & 2 & 3 & 4 \end{bmatrix}. \quad \text{Finally, do } (C_2)_{-2r_1 + r_2 \rightarrow r_2} \text{ to obtain } D_2.$$

$$D_2 = \begin{bmatrix} 1 & 1 & -\frac{5}{2} & 1 & 2 \\ 0 & 1 & \frac{3}{2} & -2 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{3}{2} & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The matrix

$$H = \begin{bmatrix} 1 & 1 & -\frac{5}{2} & 1 & 2 \\ 0 & 1 & \frac{3}{2} & -2 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{3}{2} & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

is in row echelon form and is row equivalent to A .

$$H = \begin{pmatrix} 1 & 1 & -5/2 & 1 & 2 \\ 0 & 1 & 3/2 & -2 & 1/2 \\ 0 & 0 & 1 & 3/2 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \text{ ezelon form}$$

↓

A matrisine satırca denktir.

Teorem: Sıfırdan farklı $m \times n$ tipteki her $A = (a_{ij})$ matrisi satırca (sütunca) indirgenmiş ezelon şeklindeki bir tek matrise satırca (sütunca) denktir.

Satırca indirgenmiş Eşelon Form:

Örnek: Bir önceki örnekte verilen A matrisine satırca denk olan indirgenmiş eşelon seklinde bir matris bulunuz.

$$H = \begin{pmatrix} 1 & 1 & -5/2 & 1 & 2 \\ 0 & 1 & 3/2 & -2 & 1/2 \\ 0 & 0 & 1 & 3/2 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$J_1 = H - 3/2 r_3 + r_2 \rightarrow r_2 = \begin{pmatrix} 1 & 1 & -5/2 & 1 & 2 \\ 0 & 1 & 0 & -17/4 & -5/2 \\ 0 & 0 & 1 & 3/2 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$J_2 = (J_1) - \frac{5}{2} r_3 + r_1 \rightarrow r_1 = \begin{pmatrix} 1 & 1 & 0 & 19/4 & 7 \\ 0 & 1 & 0 & -17/4 & -5/2 \\ 0 & 0 & 1 & 3/2 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$K = (J_2) - r_2 + r_1 \rightarrow r_1 = \begin{pmatrix} 1 & 0 & 0 & 9 & 19/2 \\ 0 & 1 & 0 & -17/4 & -5/2 \\ 0 & 0 & 1 & 3/2 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Satırca indirgenmiş eşelon form

2.2 Lineer Sistemin Çözümü:

Hatırlatma:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ &\vdots \\ b_{m1}x_1 + b_{m2}x_2 + \dots + b_{mn}x_n &= b_m \end{aligned} \quad (1)$$

(1) sistemi

$$\underline{A} \underline{x} = \underline{b}$$

şeklinde yazılabilir.

$(A:b)$

Bu bölümün amacı, verilen bir lineer sistemi temsil eden ilaveli matrisi; çözümü kolaylıkla bulup bir ekle dönüştürmektir. Şimdi, lineer sistemin çözümünü için satır işlemleri uygulayacağız.

Teorem 2.2.1:

$\underline{A} \underline{x} = \underline{b}$ ve $\underline{C} \underline{x} = \underline{d}$ n bilinmeyenli m lineer denklemleri iki lineer sistem olan. Eğer $(A:b)$ ve $(C:d)$ ekle matrisleri Satırca denktir bu iki lineer sistem denktir, yani sistemler tam olarak aynı çözüme sahiptir.

Sonuç: A ve C, $m \times n$ tipinde satırca denktir; matris ise

$$\underline{A} \underline{x} = \underline{0} \quad \text{ve} \quad \underline{C} \underline{x} = \underline{0}$$

hom. lineer sistemleri denktir.

Gauss Yok Etme

Adım 1. $(A:b)$ ilaveli matrisin, elemanlar satır işlemleri kullanarak satırca eşelon şeklindeki $(C:d)$ matrise dönüştürülmesi

Adım 2. Geriye yine kayma kullanarak $(C:d)$ ekli matrise karşılık gelen lineer sistemin çözümü

Gauss-Jordan Yok Etme (indijene metodu)

$(C:d)$ matrisini satırca indijene eşelon şeklinde veren metodu Gauss-Jordan indijene metodu dır.

Örnek:

$$\left. \begin{aligned} x_1 + x_2 + 3x_3 &= -2 \\ 2x_1 + x_2 - 2x_3 &= 8 \\ 3x_1 + 4x_2 + (a^2 - 3)x_3 &= 3a - 14 \end{aligned} \right\}$$

$a \in \mathbb{R}$ olm. üzere

(i) tek çözüm

(ii) sonsuz çözüm

(iii) çözüm olmaması

icn a ne olmalıdır?

Gözüm:

$$\begin{pmatrix} 1 & 1 & 3 & | & -2 \\ -2 & -1 & -2 & | & 8 \\ 3 & 4 & a^2-3 & | & 3a-14 \end{pmatrix} \xrightarrow{2R_1+R_2 \rightarrow R_2}$$

$$\begin{pmatrix} 1 & 1 & 3 & | & -2 \\ 0 & 1 & 4 & | & 4 \\ 0 & 1 & a^2-4 & | & 3a-8 \end{pmatrix} \xleftarrow{-3R_1+R_3 \rightarrow R_3} \begin{pmatrix} 1 & 1 & 3 & | & -2 \\ 0 & 1 & 4 & | & 4 \\ 3 & 4 & a^2-3 & | & 3a-14 \end{pmatrix}$$

$-R_2+R_3 \rightarrow R_3$

$$\begin{pmatrix} 1 & 1 & 3 & | & -2 \\ 0 & 1 & 4 & | & 4 \\ 0 & 0 & a^2-16 & | & 3a-12 \end{pmatrix}$$

$$(a^2-16)x_3 = 3a-12 \Rightarrow (a-4)(a+4)x_3 = 3(a-4)$$

$a=4$ için:

$$\begin{pmatrix} 1 & 1 & 3 & | & -2 \\ 0 & 1 & 4 & | & 4 \\ 0 & 0 & 0 & | & 0 \end{pmatrix} \downarrow x_2+4x_3=4$$

$$x_3 = t$$

$$x_2 = 4-4t$$

$$x_1 = -6+t$$

$$(-6+t, 4-4t, t) \in \mathbb{R}^3$$

sonuç çözüm

for $a = -4$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 3 & -2 \\ 0 & 1 & 4 & 4 \\ 0 & 0 & 0 & 24 \end{array} \right) \rightarrow 0 \cdot x_3 = 24 \quad ! \quad \begin{array}{l} \text{Çelişki} \\ \text{Çözüm Yok} \end{array}$$

$a \neq -4$:

$$(a^2 - 16) x_3 = 3a - 12 \Rightarrow x_3 = \frac{3a - 12}{a^2 - 16} = \frac{3}{a + 4}$$

$$x_2 + 4 \left(\frac{3}{a + 4} \right) = 4 \Rightarrow x_2 = 4 - \frac{12}{a + 4} = \frac{4a + 4}{a + 4}$$

$$x_1 + \left(\frac{4a + 4}{a + 4} \right) + 3 \left(\frac{3}{a + 4} \right) = -2 \Rightarrow x_1 = -2 - \frac{4a + 13}{a + 4}$$

$$\left(-2 - \frac{4a + 13}{a + 4}, \frac{4a + 4}{a + 4}, \frac{3}{a + 4} \right)$$

tek çözüm

Örnek:

$$\alpha, \beta \in \mathbb{R}$$

$$x_1 + 2x_2 - 2x_3 = -1$$

$$2x_1 + 3x_2 + (\alpha - 6)x_3 = 10$$

$$x_1 + x_2 + (\alpha^2 + \alpha - 8)x_3 = 3\alpha + 5$$

(i) sonsuz çözüm

(ii) çözüm olmaması

(iii) tek çözüm

olması için α ve β yi belirleyin.

Çözüm:

$$\left(\begin{array}{ccc|c} 1 & 2 & -2 & -1 \\ 2 & 3 & \alpha - 6 & 10 \\ 1 & 1 & \alpha^2 + \alpha - 8 & 3\alpha + 5 \end{array} \right) \begin{array}{l} -2R_1 + R_2 \rightarrow R_2 \\ -R_1 + R_3 \rightarrow R_3 \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & 2 & -2 & -1 \\ 0 & 1 & 2-\alpha & -12 \\ 0 & 0 & \alpha^2 - 4 & 3\alpha - 6 \end{array} \right) \begin{array}{l} -R_2 \rightarrow R_2 \\ -R_2 + R_3 \rightarrow R_3 \end{array} \left(\begin{array}{ccc|c} 1 & 2 & -2 & -1 \\ 0 & -1 & \alpha - 2 & 12 \\ 0 & -1 & \alpha^2 + \alpha - 6 & 3\alpha + 6 \end{array} \right)$$
$$\downarrow (\alpha^2 - 4)x_3 = 3\alpha - 6$$

i) $\alpha = 2$ için:

$$\left(\begin{array}{ccc|c} 1 & 2 & -2 & -1 \\ 0 & 1 & 0 & -12 \\ 0 & 0 & 0 & 0 \end{array} \right) \Rightarrow \begin{array}{l} x_1 + 2x_2 - 2x_3 = -1 \\ x_2 = -12 \\ x_1 - 2x_3 = 23 \\ x_2 = -12 \end{array}$$

$$\Rightarrow x_3 = t \in \mathbb{R} \text{ (parametre)}$$

$$\Rightarrow x_1 = 23 + 2t$$

$$(x_1, x_2, x_3) = (23 + 2t, -12, t) \in \mathbb{R}^3, t \in \mathbb{R}$$

Sonuç Gözetim

ii) $\alpha = -2$ için:

$$\begin{pmatrix} 1 & 2 & -2 & | & -1 \\ 0 & 1 & 4 & | & -12 \\ 0 & 0 & 0 & | & -12 \end{pmatrix}$$

$$0x_1 + 0x_2 + 0x_3 = -12$$

$0 = -12!$ Çelişki
Gözlem yok.

iii) $\alpha \neq \pm 2$ için: $3\alpha - 6 \neq 0, \alpha - 2 \neq 0, \alpha^2 - 4 \neq 0$

$$\begin{pmatrix} 1 & 2 & -2 & | & -1 \\ 0 & 1 & -(\alpha-2) & | & -12 \\ 0 & 0 & \alpha^2-4 & | & 3\alpha-6 \end{pmatrix} \xrightarrow[\alpha^2-4 \neq 0]{\frac{1}{\alpha^2-4} R_3 \rightarrow R_3} \begin{pmatrix} 1 & 2 & -2 & | & -1 \\ 0 & 1 & -(\alpha-2) & | & -12 \\ 0 & 0 & 1 & | & \frac{3}{\alpha+2} \end{pmatrix}$$

$$x_3 = \frac{3}{\alpha+2}$$

$$x_2 - (\alpha-2)x_3 = -12$$

$$x_2 - (\alpha-2)\frac{3}{\alpha+2} = -12$$

$$\Rightarrow x_2 = \frac{3\alpha - 6}{\alpha + 2} - 12$$

$$1. x_1 + 2x_2 - 2x_3 = -1 \Rightarrow x_1 = -1 + 2x_3 - 2x_2$$

$$\Rightarrow x_1 = -1 + \frac{6}{\alpha + 2} + 24 + \frac{12 - 6\alpha}{\alpha + 2}$$

$$\Rightarrow x_1 = 23 + \frac{18 - 6\alpha}{\alpha + 2}$$